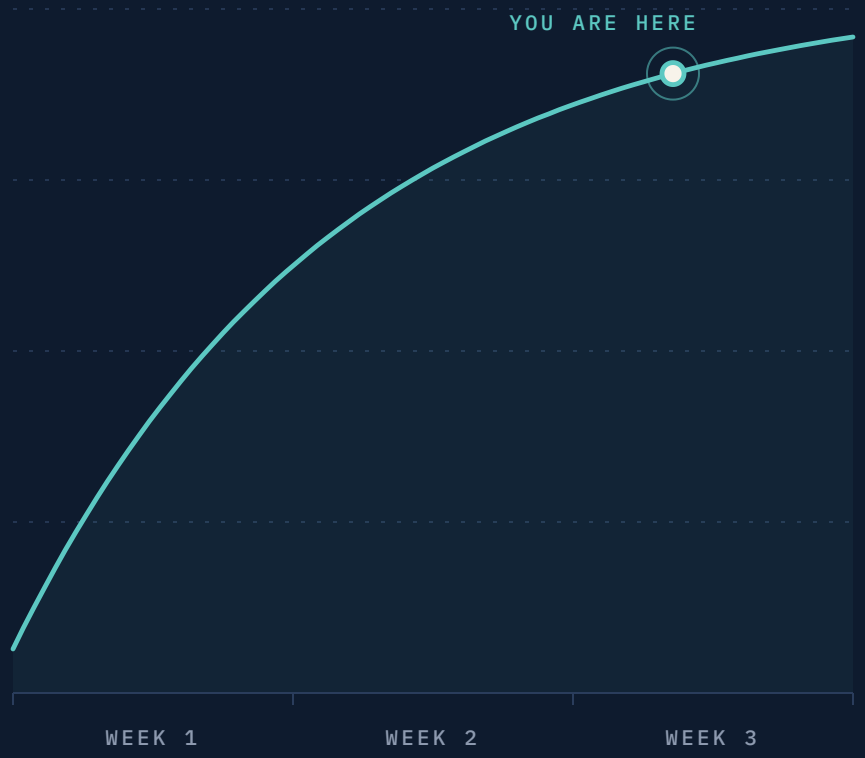
WEEK 3 · DAY 2 · OVERVIEW

Spread Over the Curve



Created by **Jeff Lenamon**

ABOUT 45 MIN WITH EXERCISES / 8 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Desk Lead's Note

Our month-end file has a column called oas. Bezane Finance's 2032 bond shows 147. Where does that come from? Rebuild it for all 200 bonds. Then tell me whether our own ten-tenor curve gives the same spreads, and what we should call the number.

ABOUT 45 MIN WITH EXERCISES / 8 SECTIONS / 4 EXERCISES

Your Route Through Day 2

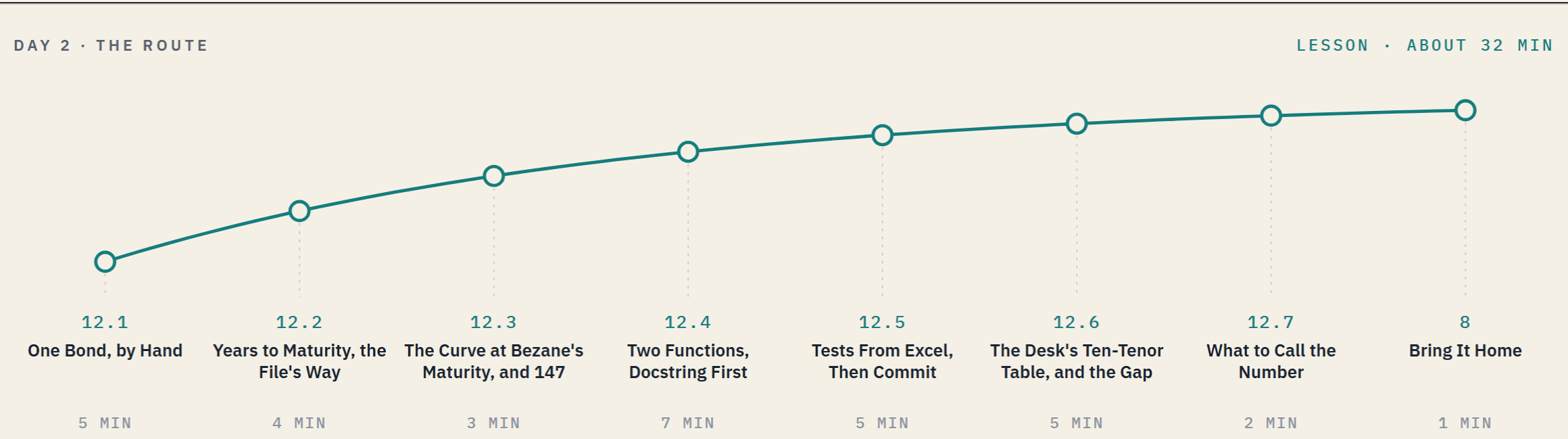


FIGURE 1 Eight sections, in order, with about how long each takes.

Time. About 32 minutes for the lesson, then about 12 for the four exercises.

12.1 One Bond, by Hand

THE BOOK

	CUSIP	ISSUER	MATURITY	YIELD_PCT	OAS
0	99080CAE8	Bezane Finance	2032-06-18	5.606	147
1	99049XAE2	Bezamyth Machinery	2029-04-17	9.220	535
2	99039NAC0	Bezoquil Freight	2030-06-03	5.509	153
3	99039NAG1	Bezoquil Freight	2053-04-09	6.756	218

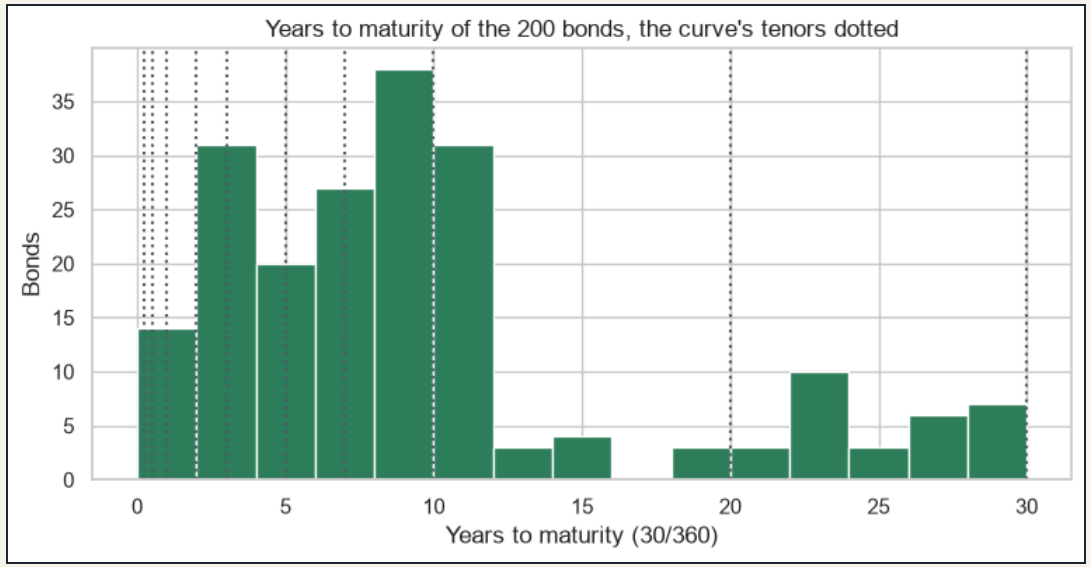
THE CURVE

```
curve = pd.read_csv('course2_treasury_curve.csv')
```

```
[0.25, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 20.0, 30.0]
[3.587, 3.622, 3.69, 3.811, 3.916, 4.086, 4.214, 4.348, 4.54, 4.586]
```

A spread to the curve is the bond's yield minus the curve's yield at the same maturity.

12.2 Years to Maturity, the File's Way



THE CODE

```
holdings['years'] = [bondmath.days_30_360(a, m) / 360
```

WHAT IT PRINTS

```
1.0056 to 29.8972 years
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Years, 30/360	`=YEARFRAC(D2,E2,0)`	5.716666666666667
The day count	`=YEARFRAC(D2,E2,0)*360`	2058

Every bond sits between two tenors, so each needs the curve read between points.

12.3 The Curve at Bezane's Maturity, and 147

THE FILE'S FORMULA

```
curve_formula_pct = round(3.55 + 1.05 * (1 - math.exp(-years / 7)), 3)
```

THE SPREAD, FROM THE FORMULA

```
print(f"{(bond['yield_pct'] - curve_formula_pct) * 100:.6f} bp; the file says {bond['oas']}")
```

```
147.000000 bp; the file says 147
```

THE STRAIGHT LINE

```
straight line: 4.131867 percent
```

EXCEL SIDE BY SIDE

JOB	EXCEL
The straight line	notebook 11's `FORECAST.LINEAR` with `INDEX` and `MATCH`
The formula	`=ROUND(3.55+1.05*(1-EXP(-G2/7)),3)`

The file read its curve from the formula at the bond's own years, rounded to 3 decimals: 147 to the last digit.

12.4 Two Functions, Docstring First

WRITE SPREAD_TO_CURVE_BP

```
%%writefile -a bondmath.py
def spread_to_curve_bp(yield_pct: float, years: float, curve_tenors: list[float],
                       curve_yields_pct: list[float]) -> float:
    """Yield spread over a curve at the bond's own maturity, in basis points.
    Units: yield_pct and curve yields in percent, years and tenors in years; the result is in bp.
    """
    curve_yield_pct = interpolate_yield(years, curve_tenors, curve_yields_pct)
    return (yield_pct - curve_yield_pct) * 100
```

THE WHOLE BOOK

```
largest miss: 2.3e-13 bp
```

Two functions rebuild the `oas` column for all 200 bonds, to 2.3e-13 of a basis point.

12.5 Tests From Excel, Then Commit

TEST A CASE WORKED BY HAND

```
%%writefile -a test_bondmath.py
def test_spread_to_curve_known_case():
    curve_at_6_years = 4.086 + (4.214 - 4.086) * (6 - 5) / (7 - 5)
    expected_bp = (5.0 - curve_at_6_years) * 100
    result = bondmath.spread_to_curve_bp(5.0, 6.0, CURVE_TENORS, CURVE_YIELDS_PCT)
    assert result == pytest.approx(expected_bp, abs=1e-9)
```

TEST THE DAY COUNT

```
def test_years_to_maturity_cases():
        (date(2026, 9, 30), date(2036, 9, 30), 3600),  # exactly ten years
        (date(2026, 9, 30), date(2032, 6, 18), 2058),  # the first bond in holdings.csv
        (date(2027, 2, 28), date(2031, 8, 31), 1621),  # the 31st end stays 31 after a 28 February start
        assert bondmath.years_to_maturity(settlement, maturity) == pytest.approx(days / 360, abs=1e-12)
```

35 tests pass, and the history says when every test arrived.

12.6 The Desk's Ten-Tenor Table, and the Gap

EVERY SPREAD ON THE TABLE

```
holdings['curve_table_pct'] = [bondmath.interpolate_yield(t, curve_tenors, curve_yields)
                               for t in holdings['years']]
holdings['spread_table_bp'] = [bondmath.spread_to_curve_bp(y, t, curve_tenors, curve_yields)
                               for y, t in zip(holdings['yield_pct'], holdings['years'])]
holdings['gap_bp'] = holdings['spread_table_bp'] - holdings['oas']
```

```
Bezane Finance on the table: 147.41 bp
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Spread to the table, bp	`=(F2-H2)*100`	147.41333333333327

The same bond on the same curve is 147 or 147.41: compare two spreads only when they share the curve, the reading between tenors, and the count of years.

12.7 What to Call the Number

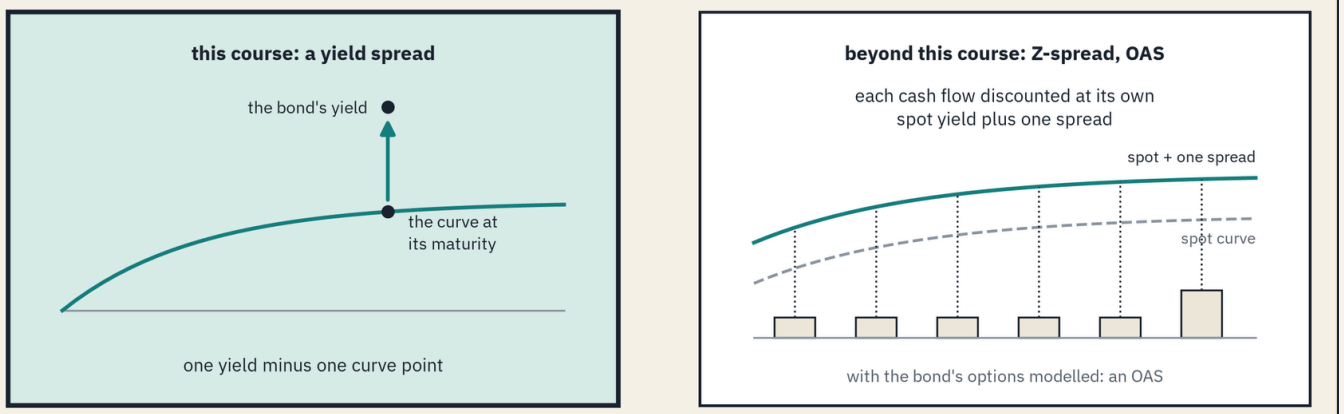


FIGURE 7 A yield spread subtracts one curve point; a Z-spread discounts each cash flow, and an OAS models options.

	OAS	SPREAD_FORMULA_BP	SPREAD_TABLE_BP
0	147	147.0	147.4133
1	535	535.0	535.1542
2	153	153.0	153.5625
3	218	218.0	218.5985
4	127	127.0	127.3713

The course calls it `spread_to_curve_bp`, a yield spread at the bond's own maturity; the file keeps the name `oas`.

Bring It Home

IN POWERSHELL

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_12_k4w7q2xr"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py", "$work\.gitignore" .
python -m pytest -q
git diff --stat
git add bondmath.py test_bondmath.py
git commit -m "Notebook 12: years_to_maturity and spread_to_curve_bp"
git log --oneline -- test_bondmath.py
git diff --stat week2
```

IF WEEK2 IS UNKNOWN

If **week2** is unknown, tag notebook 10's commit: **`git tag week2`** followed by its hash.

IN COLAB

In Colab, download the two files from the work folder in the file browser.

You Can Now

- Say what a spread to the curve measures, in bp.
- Count years to maturity on 30/360 with **`years_to_maturity`**, as Excel's **`YEARFRAC`** with basis 0 does.
- Compute a spread with **`spread_to_curve_bp`** on any curve, a one-point curve included.
- Rebuild the file's **`oas`** column from the curve's formula.
- Measure how far the reading between tenors moves a spread, and say when two spreads compare.
- Call the number a yield spread, and list the commits that changed one file.

NEXT
WEEK 3 · DAY 3

Spread Duration and DTS

Spread duration measured by moving the spread and repricing, DTS built from it, and the book's spread risk by rating bucket.